

Abdiel M. Vallejo Padró

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PROFESSIONAL SUMMARY

Industrial Engineering student at UPRM and NASA RASC-AL competitor with experience in autonomous systems, data analysis, and full-stack development. Combines hardware and software expertise to improve processes, evaluate technologies, and build data-driven solutions.

EDUCATION

University of Puerto Rico, Mayagüez (UPRM)

B.S. Industrial Engineering | GPA: 3.5

Mayagüez, PR

Expected 2029

TECHNICAL EXPERIENCE

NASA RASC-AL Competition | Lead Hardware & Autonomous Systems Engineer

Jun. 2025

- Designed a **fault tolerant autonomous computing architecture** using dual radiation hardened NASA HPSC processors and an AMD Versal AI Core, incorporating SHERPA based fault recovery to improve system resilience for lunar operations.
- Developed the rover's **autonomous ConOps workflow** by integrating system health checks, AI driven task prioritization, path planning, fault handling, and mission execution into a structured autonomy architecture.
- Optimized sensor placement** and validated perception coverage through ROS Noetic/Gazebo lunar navigation simulations on Ubuntu, supporting autonomous navigation design decisions.

SPARC, UPRM | Research Member

Aug. 2024 – Present

- Evaluated **5+ AI technologies** for lunar rover systems through **structured trade studies**, translating technical findings into **recommendations adopted by the team** for system development.

PERSONAL PROJECTS

Barber Bot | WhatsApp Booking Automation

2025 – Present

- Automated end to end appointment booking** for barbershops by developing a conversational assistant with Python, FastAPI, Groq, Redis, and the Meta WhatsApp API, reducing the need for manual appointment coordination.
- Integrated Google Calendar with the booking workflow to provide **real time availability**, appointment scheduling, reminders, and cancellations through WhatsApp.

Stock Analyzer | Open Source | Financial Analysis Platform

2026

- Built an **11 module** financial analysis backend with Python and FastAPI, exposing **13 API endpoints** for DCF, VWAP, Sharpe ratio, VaR, and SEC EDGAR insider analysis to centralize multiple investment metrics in one platform.
- Integrated yfinance, Alpha Vantage, and SEC EDGAR with caching at **zero API cost**, while enabling users to connect their own Reddit API credentials for **customizable market sentiment analysis**.

LEADERSHIP & ACTIVITIES

World Robot Olympiad | NASA Booth

Oct. 2024

- Presented the NASA RASC-AL rover and autonomous systems concepts to audiences with varying technical backgrounds, expanding accessibility of the team's engineering work during STEM outreach.

AI Winter School | University of Notre Dame

Jan. 2025

- Built and analyzed **linear regression models** using Python, Jupyter Notebook, and Google Colab during an intensive AI program, applying data analysis concepts to hands on modeling exercises.

TECHNICAL SKILLS

Programming & Data: Python, SQL, C, JavaScript, Pandas, FastAPI, React

Engineering & Tools: Power BI, NumPy, MATLAB, SolidWorks, ROS Noetic, Gazebo, Redis, Git